Radiotherapy for anal cancer

Radiotherapy is the use of high-energy rays to treat cancer. It can be given on its own or with chemotherapy to treat anal cancer.

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Anal cancer radiotherapy

Radiotherapy uses high-energy rays to destroy cancer cells. It destroys cancer cells in the area of the body you have it, while doing as little harm as possible to normal cells.

Some normal cells in the area being treated can also be damaged by radiotherapy. This can cause side effects. As the normal cells recover, the side effects usually get better.

Radiotherapy is always carefully planned by a team of experts. They will plan your treatment so it does as little harm as possible to normal cells.

For anal cancer, you usually have radiotherapy in combination with chemotherapy. This is called chemoradiation. You may have radiotherapy on its own if your doctors think you are not well enough to have chemoradiation.

Radiotherapy may also be used to relieve symptoms if the cancer has spread to other parts of the body.

Radiotherapy does not make you radioactive. It is safe for you to be with other people throughout your treatment, including children.

Getting support

We understand that having treatment can be a difficult time for people. We're here to support you. If you want to talk, you can:

- Call the Macmillan Support Line for free on 0808 808 00 00.
- Chat to our specialists online.
- Visit our anal cancer forum to talk with people who have been affected by anal cancer, share your experience, and ask questions.

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External beam radiotherapy

External beam radiotherapy is given from a radiotherapy machine. The machine looks like a large x-ray machine or CT scanner.

You usually have a number of short, daily treatments in a radiotherapy department. These are called treatment sessions or fractions.

Each treatment takes 10 to 15 minutes. The treatments are usually given Monday to Friday, with a break at the weekend. Your doctor will talk with you about the treatment and possible side effects.

Intensity-modulated radiotherapy (IMRT)

A type of radiotherapy called IMRT is usually used for anal cancer.

IMRT shapes the radiotherapy beams. This allows different doses of radiotherapy to be given to different parts of the treated area. It means lower doses of radiotherapy are given to the healthy tissue surrounding the tumour. This can help reduce the risk of side effects and late effects. It may also allow higher doses of radiotherapy to be given to the tumour.

Your doctor can tell you more about IMRT and whether it is a suitable treatment for you.

Volumetric-modulated arc radiotherapy (VMAT)

VMAT is a newer way of giving IMRT. It is sometimes called RapidArc®. The radiotherapy machine moves around you and reshapes the beam during treatment. This makes it more accurate and shortens the treatment time.

Planning your radiotherapy treatment

You will have a hospital appointment to plan your treatment. You will usually have a CT scan of the area to be treated. During the scan, you need to lie in the position that you will be in for your radiotherapy treatment.

Your radiotherapy team use information from this scan to plan:

- the dose of radiotherapy
- the area to be treated.

You may have some small, permanent markings made on your skin. The marks are about the size of a pinpoint. They help the radiographer make sure you are in the correct position for each session of radiotherapy.

These marks will only be made with your permission. If you are worried about them, tell your radiographer.

The doctor may place a small metal marker on the skin around your anus. The marker shows up on the scan so the doctor can see the exact area to be treated.

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Preparing for your radiotherapy

You will be sent information about having a full bladder for your treatment sessions as guidance to when you should start drinking fluids. This will also be discussed during your first treatment session. The advice may change during your course of treatment, as your ability to hold urine will change.

Treatment sessions

The radiographer will explain what will happen. At the start of each treatment session (called a fraction), they will make sure you are in the correct position on the couch and that you are comfortable.

When everything is ready, they leave the room so you can have the radiotherapy. The treatment only takes a few minutes. You can talk to the radiographers through an intercom or signal to them during the treatment. They can see and hear you from the next room.

During treatment, the radiotherapy machine may automatically stop and move into a new position. This is so the radiotherapy can be given from different directions.

Side effects of radiotherapy

You may have side effects during your treatment. Side effects build up slowly when you start treatment. They usually get better slowly over a few weeks or months after treatment finishes.

Your cancer doctor, nurse or radiographer will discuss this with you so you know what to expect. Tell them about any side effects you have during or after treatment. There are often things they can do to help.

We have more information about managing side effects of pelvic radiotherapy.

Side effects caused by smoking

The side effects of radiotherapy are made worse by smoking. Smoking will also make your treatment less effective. Your cancer doctor or nurse will advise you to try to stop smoking. They can give you support and advice.

Tiredness

Tiredness (fatigue) can continue for weeks or a few months after your treatment has finished. You might be more tired if you have to travel to hospital each day. If you are also having other treatment such as chemotherapy, this can make you even more tired.

Try to get as much rest as you can, especially if you have to travel a long way for treatment. Balance this with some physical activity, such as short walks, which will give your more energy.

Skin reaction

This is a common side effect. The skin in the area that is treated, such as around the anus, the groin, vagina or scrotum (where the testicles are) may:

- feel sore or itchy

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- redden if you have white skin
- become darker than surrounding skin if you have black or brown skin
- become moist and blister or peel – which can be painful towards the end of treatment.

Your radiographer or specialist nurse will give you advice about looking after your skin. Tell them straight away about any skin changes. They can give you advice or treatments such as creams or dressings. They can also prescribe painkillers for you.

During treatment, you are usually advised to:

- wash your skin gently with mild, unperfumed soap and water
- gently pat dry and avoid rubbing your skin
- wear loose-fitting clothes made from natural fibres, such as cotton
- follow your radiotherapy team's advice about moisturising the treated area
- spend some time sitting or lying on your side to help to relieve any discomfort around the back passage and perineum.

Skin reactions should start getting better within 4 weeks of treatment finishing.

Pain

Skin reactions can be painful especially when you pass stools (poo). Your doctor can prescribe local anaesthetic creams to help with this. But you might need stronger painkillers to take regularly until the pain improves. Your doctor will prescribe these for you. Always tell your doctor, nurse or radiographer if you have any pain.

Bowel changes

You may experience problems with your bowel such as:

- loose stools (poo)
- passing a lot of wind
- needing to pass stools urgently and more frequently
- cramping pains in your tummy or back passage
- constipation
- tenesmus – this is the feeling that you need to go to the toilet even if your bowel is empty
- having some bleeding or mucus coming from the back passage.

If you have diarrhoea, your doctor will prescribe anti-diarrhoea tablets to help. Tell your specialist nurse or radiographer if you have any leakage (incontinence). They will give you advice and explain how to look after the skin in that area.

You might be advised to make changes to your diet during radiotherapy.

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Inflammation of the bladder (cystitis)

Radiotherapy to the anal area may cause inflammation of the lining of the bladder. This can make you feel you want to pass urine more often. It also gives you a burning feeling when you pass urine.

It helps to drink plenty of water and other fluids. Your doctor can prescribe medicine to make passing urine more comfortable.

Feeling sick

Radiotherapy may make you feel sick (nausea) or be sick (vomit). Sickness can usually be well controlled and stops when treatment finishes. Your healthcare team may give you anti-sickness (anti-emetic) drugs to prevent nausea or vomiting.

Hair loss

Most people lose their pubic hair. It should grow back after your treatment finishes, although the hair loss is sometimes permanent.

Changes to the vagina

Radiotherapy can make the lining of the vagina sore and inflamed. You may be advised not to have sex during treatment and for a few weeks after.

After radiotherapy, the vagina may be narrower, less stretchy and drier than before. This may make having sex painful or make it difficult if you need to have internal examinations. Your nurse may recommend you use vaginal dilators to try to prevent the vagina from narrowing. Dilators are plastic tubes shaped like tampons. They are available in different sizes that you use with a lubricant.

Vaginal dryness can be relieved with vaginal lubricants or creams. Hormone creams can also help with dryness and vaginal narrowing. Your doctor can prescribe these.

Early menopause

If you are still having periods, radiotherapy to the pelvic area may cause the menopause. This means you will not be able to get pregnant. Your doctor can prescribe hormone replacement therapy (HRT) to improve symptoms of the menopause.

Effects on the ovaries and womb

Radiotherapy for anal cancer can affect the ovaries and the lining of the womb. This often means you will not be able to get pregnant (infertility) or carry a pregnancy after treatment. Your team will give you information about this.

If you would like to have children in the future, talk to your doctor or specialist nurse before you start treatment. There may be options for preserving your fertility.

Contraception

It is important that you do not get pregnant or make someone pregnant during radiotherapy, and for a few months after it has finished. Your doctors usually recommend that you use contraception during this time.

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Problems getting an erection

Radiotherapy for rectal cancer can cause problems getting or keeping an erection. This is called erectile dysfunction or ED. Your radiotherapy team can explain what is likely to happen. There are different treatments that can help with ED.

You may also have a sharp pain when you ejaculate. The pain should get better a few weeks after treatment finishes. You might also find that you ejaculate less fluid or none.

Effects on sperm

Radiotherapy to the pelvis might make you unable to make someone pregnant (infertile). It may be possible to have sperm stored before treatment starts. This is called sperm banking. It is important to talk to your cancer doctor or specialist nurse about this before your treatment starts.

Booklets and resources

Booklets

01 Mar 2024

Cancer and your sex life

Booklets

01 Mar 2024

Your sex life and cancer - easy read

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Possible late effects of radiotherapy

You may still have side effects months after treatment finishes. Or you may develop new side effects months or years later. These are called long-term side effects or late effects.

Newer ways of giving radiotherapy aim to reduce the risk of getting late effects. There is slightly more risk of developing late effects when you have radiotherapy and chemotherapy together. Your doctor or nurse will talk to you about this.

Some late effects include:

- bladder changes, such as needing to pass urine more often or urgently
- bowel changes, such as diarrhoea or bleeding from the back passage
- effects on your sex life.

We have more information about these and other late effects of pelvic radiotherapy. This includes advice on how they can be managed or treated.

About our information

This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer.

References

Below is a sample of the sources used in our anal cancer information. If you would like more information about the sources we use, please contact us at informationproductionteam@macmillan.org.uk

S Rao, MG Guren, K Khan et al. Anal cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. 2021. European Society of Medical Oncology. Available from: www.esmo.org/guidelines/guidelines-by-topic/esmo-clinical-practice-guidelines-gastrointestinal-cancers/anal-cancer [accessed January 2023].

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Our cancer information meets the PIF TICK quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about how we produce our information.

The language we use

We want everyone affected by cancer to feel our information is written for them.

We want our information to be as clear as possible. To do this, we try to:

- use plain English
- explain medical words
- use short sentences
- use illustrations to explain text
- structure the information clearly
- make sure important points are clear.

We use gender-inclusive language and talk to our readers as ‘you’ so that everyone feels included. Where clinically necessary we use the terms ‘men’ and ‘women’ or ‘male’ and ‘female’. For example, we do so when talking about parts of the body or mentioning statistics or research about who is affected.

You can read more about how we produce our information here.

How we can help

Clinical Information Nurse Specialists

Our Cancer Information Nurse Specialists are dedicated cancer nurses available to talk to on our Macmillan Cancer Support Line.

Call us free on 0808 808 0000

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8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.

Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don’t want to call.

Anal cancer forum

Join our anal cancer forum and chat anonymously to others who understand what you are going through. Our community is available 24/7. It is a safe space where you can share experiences, ask questions and talk to people who understand.

Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

Get help

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